

Notes: Goldstein, Spatt, and Ye (RFS, 2021)

Big Data in Finance

Contents

1	Big picture	3
2	Definitions, data, and measurement (key objects)	4
2.1	Why finance needs its own definition of big data	4
2.2	Large size	4
2.3	High dimension	4
2.4	Complex structure	5
2.5	How the six papers map into the taxonomy	5
3	Models and empirical specifications (none in the usual sense)	6
4	What makes the article persuasive	6
5	The six papers in the special issue	7
5.1	Erel et al. (2021): selecting directors using machine learning	7
5.2	Li et al. (2021): measuring corporate culture using machine learning	7
5.3	Easley et al. (2021): microstructure in the machine age	8
5.4	Anand et al. (2021): institutional order handling and broker-affiliated venues	9
5.5	Benamar, Foucault, and Vega (2021): clickstream demand for information	9
5.6	Giglio, Liao, and Xiu (2021): thousands of alpha tests	10
6	Identification and instruments (what makes the arguments credible here)	10
7	Where does big-data finance go from here?	11
7.1	4.1 Machine learning and learning machines	11
7.2	4.2 Feedback effects of the big data revolution	11
7.3	4.3 Heterogeneous impact of the big data revolution	12
7.4	4.4 More complex data	12

7.5	4.5 Regulations	12
7.6	4.6 Theory	13
7.7	4.7 Interdisciplinary collaborations	13
8	Main points	13
9	Short summary	14
10	Conclusion	14

1 Big picture

- **What kind of paper is this?** This is an *introduction to a special issue*, not a stand-alone empirical or theoretical paper with one dataset, one regression table, or one structural model. Its job is to define what “big data” means in finance, show how six papers in the special issue fit that theme, and lay out a research agenda.
- **Main question.** What should count as *big data in finance research*, and how can it change the way financial economists study asset pricing, corporate finance, and market microstructure?
- **Core conceptual contribution.** The paper argues that the engineering-style definition of big data as *volume, velocity, and variety* is not quite the right one for finance. Instead, it proposes a finance-specific definition based on **three properties: large size, high dimension, and complex structure**.
- **Why this matters.** In finance, the key issue is not just how to store or process data. The key issue is whether new data and new tools help answer economic questions that could not be answered well with “small data.”
- **How the paper proceeds.**
 - define the three properties of big data in finance,
 - map the six special-issue papers into that framework,
 - explain what each paper contributes,
 - and sketch where research should go next.
- **Main message.** Big data in finance is not simply “more observations.” It also includes situations where the data are high-dimensional, where the relevant information is unstructured, and where machine-learning tools become useful because interactions, nonlinearities, or prediction tasks are central.
- **Methodological message.** Machine learning is not treated as magic. The paper is careful to say it is most economically useful when:
 - the problem genuinely involves many variables,
 - nonlinearities or interaction effects matter,
 - and prediction is more important than classical statistical inference.
- **Substantive message.** Big data techniques are already affecting very different finance questions:
 - board selection and corporate culture in corporate finance,
 - market quality and broker conflicts in market microstructure,
 - and multiple testing and false discoveries in asset pricing.
- **Policy message.** As machines become important decision-makers in finance, existing rules designed for humans may become outdated. Big data therefore has a regulatory, privacy, and fairness dimension in addition to a purely econometric one.
- **What the paper is *not*.** It does *not* offer its own causal design, estimation strategy, or structural model. Its contribution is conceptual and agenda-setting.

- **Economic takeaway.** The frontier of finance research is increasingly about combining economic questions with new data sources, flexible statistical tools, and sometimes collaboration with computer science, statistics, and large-scale computing infrastructure.

2 Definitions, data, and measurement (key objects)

2.1 Why finance needs its own definition of big data

- The paper starts from the observation that standard big-data definitions were built mainly for engineering and statistics, where the focus is on storage, processing, and infrastructure.
- Financial economists care about a different problem: whether new data and new methods help answer *economic* questions in a more convincing way.
- The authors therefore define big data in finance using **three features taken together: large size, high dimension, and complex structure**.
- Their definition is intentionally broad and provisional. They explicitly present it as a starting point that later research can refine.

2.2 Large size

- **Definition.** Data are big when they are large in an **absolute** or **relative** sense.
- **Absolute size.** A natural example is transaction-level market microstructure data. The paper notes that one day of current option trading data alone is roughly **two terabytes**.
- **Relative size.** A dataset can also be “big” relative to the best existing dataset in the literature. Many “small” datasets are just subsets of a larger universe, created by aggregation, subsampling, or periodic snapshots.
- **Why relative size matters.** The larger underlying dataset can be economically important when it:
 - corrects sample-selection bias,
 - avoids attrition from self-reported data,
 - or captures economically important activity that aggregated data miss.
- **Key intuition.** In finance, “bigger” is useful when it changes identification, measurement, or the economic phenomenon one can observe—not merely because it contains more rows.

2.3 High dimension

- **Definition.** High-dimensional data contain many variables relative to the sample size, or more broadly involve a very rich function class because interactions and nonlinearities matter.
- **Why machine learning enters.** The paper presents machine learning as a common response to the high-dimension challenge.

- **When machine learning is economically meaningful.** The authors say machine-learning methods are especially useful when:
 1. the actual problem involves lots of variables,
 2. the impact of variables is highly nonlinear or depends on interactions,
 3. and prediction is more economically central than classical inference.
- **Natural setting.** These conditions arise especially when the decision-makers themselves are machines, such as algorithmic traders or robo-advisors.
- **Important nuance.** High dimension can come from different sources:
 - many possible covariates and interaction terms,
 - many hypothesis tests, as in factor-zoo questions,
 - or a prediction task where functional-form flexibility matters more than coefficient-by-coefficient interpretation.

2.4 Complex structure

- **Definition.** Data are complex when they are not in the traditional row-column format.
- **Examples.** Text, pictures, videos, audio, and voice all count as complex or unstructured data.
- **Why this matters economically.** These data create value when they measure economic activity that structured data cannot capture well.
- **First step in practice.** Researchers usually have to extract features from the raw data before doing economics.
- **Typical tools.**
 - **NLP** for semantic information in text,
 - **speech recognition** for tone or voice information,
 - **computer vision** for geographic, facial, or image-based information.
- **Important connection.** Unstructured data are often high-dimensional by nature, so the “complex structure” and “high dimension” categories frequently overlap.

2.5 How the six papers map into the taxonomy

- **Large size.**
 - Anand et al. (2021): OATS data with broker identities and no self-reporting attrition.
 - Easley et al. (2021): large market microstructure data.
 - Benamar, Foucault, and Vega (2021): large clickstream data around macro news.
- **High dimension.**
 - Erel et al. (2021): nonlinearities and interaction terms in director selection.
 - Easley et al. (2021): high-dimensional prediction in cross-asset microstructure.

- Giglio, Liao, and Xiu (2021): many simultaneous alpha tests and false-discovery problems.
- **Complex structure.**
 - Li et al. (2021): text from earnings calls transformed into culture measures.
 - Benamar, Foucault, and Vega (2021): clickstream traces, obtained via a vendor that converts raw behavioral data into structured measures.
- **Cross-field scope.** The point of the taxonomy is that big data is not confined to one subfield. The six papers span **corporate finance, market microstructure, and asset pricing**.

3 Models and empirical specifications (none in the usual sense)

- **No standalone model.** This article does not present a new formal model, no regression equation, and no structural estimation problem of its own.
- **No single dataset.** There is no new sample assembled by Goldstein, Spatt, and Ye. Instead, the paper synthesizes six distinct papers and explains what each contributes.
- **What replaces a model here?** The article’s core “specification” is conceptual:
 1. define the object *big data in finance*,
 2. decompose it into three characteristics,
 3. illustrate each characteristic using concrete papers,
 4. and then derive a forward-looking research agenda.
- **Why this is still useful.** For a survey-style introduction, the contribution is not identification but **organization**. The paper gives a framework that helps a reader understand how very different empirical designs belong to one broader methodological shift.
- **Important methodological lesson.** The paper repeatedly stresses that machine learning is useful only when the economic problem justifies it. It does not claim that finance should blindly replace standard econometrics with black-box methods.
- **Key example of this restraint.** In the Easley et al. paper discussed below, simple logistic regression performs almost as well as more complex machine-learning methods for within-asset predictions, while machine learning clearly helps for cross-asset predictions. The message is: *use complexity when the economic setting requires it*.

4 What makes the article persuasive

- **It defines, then shows.** The paper first proposes a taxonomy and then immediately demonstrates that the taxonomy has content by mapping six papers into it.
- **It is broad but concrete.** Rather than staying at a slogan level, the paper gives specific examples from trading data, clickstream data, board-selection problems, text analysis, and asset-pricing multiple testing.

- **It does not oversell machine learning.** The paper highlights places where machine learning helps, but also emphasizes that simple models can still perform well when the underlying economics are already well understood.
- **It links methods to economics.** The three categories are not framed as purely technical data traits. Each category is tied to what new economic question can be asked or answered more credibly.
- **It incorporates policy.** The article does not stop at empirical technique. It also points to regulation, data ownership, privacy, and fairness as core future issues.
- **It is honest about limits.** The authors present their definition as provisional and acknowledge that the meaning of big data in finance will evolve.

5 The six papers in the special issue

5.1 Erel et al. (2021): selecting directors using machine learning

- **Question.** Can machine learning do better than managers when firms choose new board members?
- **Why this is a big-data problem.** The challenge is not raw sample size so much as **high-dimensional interactions and nonlinearities**. The determinants of director quality may depend on complex combinations of observable characteristics.
- **Main result.** Directors whom the algorithm predicts will perform poorly do indeed perform poorly relative to realistic candidate pools in out-of-sample tests.
- **How performance is measured.** The paper notes that measuring individual director performance is difficult because boards act collectively. The cited study therefore uses shareholder support in annual reelections as its main metric.
- **Additional findings highlighted in the introduction.** Relative to algorithm-selected directors, management-selected directors who later receive predictably low shareholder approval are more likely to:
 - be male,
 - have larger networks,
 - and sit on more boards.
- **Interpretation.** Human nomination decisions may reflect homophily, bias, or agency conflicts. The algorithm may select a more diverse set of directors.
- **Why it matters.** The paper is presented as one of the early corporate-finance applications showing that machine learning can uncover patterns that standard OLS misses because interactions and nonlinearities matter.

5.2 Li et al. (2021): measuring corporate culture using machine learning

- **Question.** Can researchers measure corporate culture using unstructured textual data rather than crude proxies?

- **Why this is a big-data problem.** The core challenge is **complex structure**: corporate culture is difficult to measure with conventional structured variables, but relevant signals may exist in text.
- **Data input.** Earnings-call transcripts.
- **Method.** A semi-supervised machine-learning approach with **word embeddings**, rather than a simple bag-of-words count.
- **Why bag-of-words is insufficient.** Counting positive and negative words may help with tone, but it does not capture the richer semantic content needed to infer culture.
- **Main conceptual output.** The paper decomposes corporate culture into five dimensions:
 - innovation,
 - integrity,
 - quality,
 - respect,
 - teamwork.
- **Applications emphasized in the introduction.** Firms scoring high on innovation and respect are more likely to be acquirers, and firms that are closer in cultural value are more likely to merge.
- **Why it matters.** This is a clean example of how unstructured data can make an otherwise elusive economic object measurable.

5.3 Easley et al. (2021): microstructure in the machine age

- **Question.** Do classical market-microstructure measures still predict price and liquidity once trading is dominated by machines and high-frequency traders?
- **Measures studied.** The introduction lists six extant measures:
 - the Roll measure,
 - Roll impact,
 - VIX,
 - Kyle’s λ ,
 - the Amihud measure,
 - VPIN.
- **Why this is a big-data problem.** The setting uses large market data, but the especially important challenge is **high-dimensional prediction**, particularly across assets.
- **Main result.** The old measures still predict future price and liquidity outcomes even in the machine age.
- **Important nuance.** The needed functional form depends on the task:
 - for **within-asset** predictions, a simple logistic regression performs almost as well as more complex machine-learning methods;

- for **cross-asset** predictions, machine learning strictly dominates simple logistic regression.
- **Interpretation.** When economists already understand the market structure of a single asset well, simple models may be enough. When cross-asset interactions are important and poorly understood, flexible methods become more valuable.
- **Why it matters.** This paper is one of the clearest examples in the introduction of machine learning being valuable because the economic environment is genuinely high-dimensional.

5.4 Anand et al. (2021): institutional order handling and broker-affiliated venues

- **Question.** Do brokers route institutional orders in ways that benefit their own affiliated trading venues at the expense of clients?
- **Why this is a big-data problem.** The central gain comes from **large size in a relative sense**. The OATS regulatory data contain broker identities and avoid attrition and selection problems that arise in self-reported data.
- **Key data source.** FINRA’s **Order Audit Trail System (OATS)**.
- **Main result.** Brokers that route more orders to their affiliated ATSs provide worse execution quality for customers, including lower fill rates and higher implementation shortfall costs.
- **Interpretation.** Brokers capture private benefits by steering flow toward affiliated venues, even when doing so is not consistent with best execution for customers.
- **Causal angle emphasized by the introduction.** Because the dataset is so large, the authors can still retain strong statistical power in a subsample and use the **SEC Tick Size Pilot** as a controlled experiment.
- **Identification result summarized by the introduction.** In a triple-difference design, execution quality improves for treated stocks when orders are handled by brokers that prefer affiliated ATSs, because the pilot constrains routing to those venues.
- **Why it matters.** The paper is a good example of large administrative data improving both measurement and causal design.

5.5 Benamar, Foucault, and Vega (2021): clickstream demand for information

- **Question.** Can online information demand measured from clickstream data proxy for uncertainty in financial markets?
- **Why this is a big-data problem.** This paper combines **large size** with **complex structure**. The raw behavior is unstructured, but a data vendor transforms it into usable structured measures.
- **Data source.** Bitly clickstream statistics from shortened URLs pointing to relevant news.
- **Economic idea.** When uncertainty about an asset rises, investors search for more information. Higher information demand should therefore predict stronger yield sensitivity to macro news.

- **Main quantitative result.** A **one-standard-deviation increase** in Bitly clicks on news related to nonfarm payroll in the **two hours before the announcement** raises the sensitivity of U.S. Treasury note yields by about **4 to 6 basis points**, depending on maturity.
- **Why that is economically large.** The paper notes that the unconditional sensitivity of Treasury yields to nonfarm-payroll announcements in the sample is only about **3.5 to 7 basis points**.
- **Timing result.** Most of the predictive content comes from clicks within the two hours immediately before the announcement.
- **Why it matters.** The paper illustrates how vendor-processed digital traces can produce new high-frequency proxies for latent economic concepts such as uncertainty.

5.6 Giglio, Liao, and Xiu (2021): thousands of alpha tests

- **Question.** How should researchers deal with data snooping and false discoveries when they test very many asset-pricing hypotheses at once?
- **Why this is a big-data problem.** The challenge is **high dimension**, but not because of raw sample size. It comes from the number of simultaneous tests in environments like the factor zoo or large mutual-fund/hedge-fund universes.
- **Core problem.** With many tested factors or funds, a large number of positive results will arise purely by chance, generating a high false discovery rate.
- **Methodological solution highlighted in the introduction.** The paper combines:
 - matrix completion,
 - wild bootstrap,
 - screening,
 - and false-discovery control.
- **Role of machine learning.** Matrix completion helps interpolate missing data and latent factors, and the latent factors help correct correlation among alpha test statistics.
- **Application.** The introduction notes a hedge-fund dataset as the paper’s illustration, while emphasizing that the toolbox is broader than this one application.
- **Why it matters.** This paper shows that in finance, big data can also mean a large model space and a large testing problem—not just a huge transactional dataset.

6 Identification and instruments (what makes the arguments credible here)

- **For this introduction itself.** The article’s credibility does not come from a single instrument or natural experiment. It comes from whether the proposed taxonomy is useful and whether the six papers genuinely fit it.
- **Why the taxonomy is credible.** The three-part classification is persuasive because each category changes what a researcher can do:

- large size changes coverage and selection,
 - high dimension changes prediction and functional-form flexibility,
 - complex structure changes measurement by turning unstructured signals into economic variables.
- **Why the article avoids overreach.** It does not claim that every finance problem is now a machine-learning problem. Instead, it explains *when* flexible methods help.
 - **One especially useful lesson.** Large data can improve causal work indirectly. Anand et al. are the best example in the introduction: because the underlying sample is so rich, the researchers can move to a sharper treated subsample and still keep enough power for a triple-difference design.
 - **Another useful lesson.** Big data can improve measurement without necessarily generating identification by itself. Li et al. and Benamar et al. mostly illustrate new measurement of latent objects such as culture or uncertainty.
 - **Main limitation of the introduction.** Since this is a survey-style essay, it cannot fully evaluate each paper’s identification strategy or external validity in the space available.
 - **How to read it well.** Treat it as a roadmap. The contribution is to tell you what kinds of problems belong under the “big data in finance” umbrella and what kinds of methodological gains each type may deliver.

7 Where does big-data finance go from here?

7.1 4.1 Machine learning and learning machines

- Most current work, including the papers in this special issue, uses machine learning to understand *human* behavior.
- The authors argue that one especially promising frontier is when the *decision-makers are machines themselves*.
- They point out a gap in the literature: much academic asset-pricing work studies monthly returns or quarterly holdings, while high-frequency research studies subsecond trading. In between lies a large world of machine-driven traders operating at horizons shorter than a month but slower than HFT.
- This middle territory is underexplored and naturally suited to machine-learning methods.
- The authors suggest that just as psychology helped generate behavioral finance, the study of algorithmic behavior could generate something like **algorithmic behavioral finance**.

7.2 4.2 Feedback effects of the big data revolution

- If investors increasingly rely on machine learning, firms may adapt their behavior to those machine readers.
- The introduction mentions work showing that firms adjust their 10-Ks and 10-Qs to cater to machine readers.

- The deeper question is whether firms will also change *real decisions*: project choice, disclosure policy, and the incentive to learn from prices.
- This is a direct extension of classic finance questions about the feedback effects from financial markets to corporate policy.

7.3 4.3 Heterogeneous impact of the big data revolution

- More data do not automatically improve welfare for everyone.
- The paper points to social media as an example where information diffusion can push prices away from fundamentals.
- The introduction uses the January 2021 GameStop episode as a vivid case where coordination through social media generated extreme volatility and major losses for a hedge fund.
- The broader research question is distributional: how does big data affect retail traders, sophisticated intermediaries, firms, and society overall?

7.4 4.4 More complex data

- The authors expect finance to move beyond text toward audio, video, and images.
- Their key point is comparative: even text transcripts are “small data” relative to the raw audio that created them.
- Audio can contain incremental information, such as managerial vocal cues, beyond the literal text.
- Satellite images and other alternative data are presented as examples of how finance may keep moving toward richer and harder-to-process signals.

7.5 4.5 Regulations

- Regulations were largely built for markets dominated by human actors. The paper argues that this may become increasingly inadequate.
- Their odd-lot example is instructive: trades below 100 shares were once treated as retail noise and excluded from reporting requirements, but algorithms later used odd lots strategically.
- The authors therefore suggest that regulation should pay more attention to the *algorithms themselves*, not only to realized transactions.
- They also highlight privacy and fairness questions:
 - who owns data,
 - what data are worth,
 - who has access,
 - and how unfair access affects outcomes.

7.6 4.6 Theory

- The special issue is mostly empirical, but the authors explicitly argue that theory is essential.
- One role for theory is to explain why interactions and nonlinearities matter—that is, why one variable’s effect depends on the level of another variable or on its own magnitude.
- Another role is to understand machine-based environments where decisions may be more internally consistent than human decisions.
- The introduction suggests that theory may fit better in some algorithmic settings precisely because machines implement rules more consistently than humans do.

7.7 4.7 Interdisciplinary collaborations

- The authors expect more collaboration with other fields.
- **Large-size challenge:** work with supercomputing centers and large-scale computing infrastructure.
- **High-dimension and complex-structure challenges:** work with scholars in math, statistics, and computer science.
- **Practical route:** researchers can also work with data vendors that provide alternative datasets such as satellite images, sentiment, or credit-card activity.
- **Big-picture lesson.** The future of big-data finance is probably institutional as well as intellectual: it requires new collaborations, infrastructure, and funding models.

8 Main points

- **The three-part definition: large size, high dimension, complex structure.**
- **Absolute versus relative size.** This distinction is one of the article’s most useful ideas. “Big” can mean huge in levels, but it can also mean a broader and less biased version of what the literature has previously used.
- **Machine learning is conditional, not universal.** It is most valuable when the economic problem is nonlinear, interaction-heavy, and prediction-oriented.
- **One memorable scale fact.** One day of option trading data can be roughly **two terabytes**.
- **One memorable measurement result.** Bitly clicks before nonfarm-payroll announcements add a large amount of explanatory power for Treasury-yield responses, with a **4–6 bps** increase in sensitivity from a one-standard-deviation increase in preannouncement clicks.
- **One memorable methodological result.** In microstructure, simple logistic models can still work well within an asset, but machine learning becomes much more useful when predicting *across* assets.
- **One memorable policy point.** Big data in finance is not just an econometrics story; it is also a story about regulation, data ownership, privacy, and fairness.

9 Short summary

- The paper defines big data in finance using three features: **large size**, **high dimension**, and **complex structure**.
- It uses six papers in the special issue to show that these features matter in corporate finance, market microstructure, and asset pricing.
- The article’s methodological point is that machine learning is useful when the economic problem is genuinely nonlinear, interaction-rich, or prediction-focused—not simply because data happen to be large.
- The article’s forward-looking point is that big-data finance will increasingly involve machines as decision-makers, more complex datasets, new regulatory issues, more theory, and stronger interdisciplinary collaboration.

10 Conclusion

- **Core conclusion.** Goldstein, Spatt, and Ye argue that the big-data revolution in finance should be understood not as a vague slogan but as a specific combination of larger datasets, richer state spaces, and more complex information sources.
- **Economic intuition.** Finance research advances when new data let us observe previously hidden actions, when flexible methods uncover nonlinear economic relations, and when unstructured information can be converted into measurable variables.
- **Takeaway.** Read this article as a roadmap for the field. The most important thing to retain is the authors’ taxonomy and the idea that big data becomes economically meaningful only when it helps answer substantive finance questions better than earlier data and methods could.